ktech
kuwait technical college

Your First Open-Source Contribution

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Using **VS Code**, the **Terminal**, and **GitHub Copilot** to Contribute to a Real Project

Lesson Overview

Duration	2 – 2.5 hours
Audience	Complete beginners — no prior experience required
OS	Windows (with macOS notes)
End Goal	Add a cybersecurity zen koan & submit a Pull Request
Repository	cybersecurity_zen_koans_app
🎥 **Video**	Full tutorial on YouTube

What You Will Learn

- 🖥️ How to use **VS Code** (editor, terminal, extensions)
- ⌨️ Basic **terminal / command-line** commands
- 🔀 **Git** fundamentals (clone, add, commit, push)
- 🌐 **GitHub** workflow (fork, pull request)
- 🤖 Using **AI (GitHub Copilot)** to assist development

The App: Cybersecurity Zen Koans

- 📱 A **real mobile app** built with Flutter – available on the **App Store** and **Google Play Store**
- 🧘 Displays short, zen-style cybersecurity sayings with technical explanations
- 📄 All koans live in a single file: `src/assets/koans.json`
- 🌍 **Anyone** can contribute new koans via a Pull Request

> Once your PR is approved and merged, a **new version of the app** will be published to both stores – **with your koan inside it**. Your words, in the hands of real users worldwide.

What is Open Source?

- 🔓 Source code is **publicly available** – anyone can view, use, modify, and share it
- 🌍 Built in the open – not locked inside one company
- 💻 **Linux**, **Android**, **Firefox**, **VS Code**, **Python** – all open source

> Think of it like a community recipe book. Anyone can read the recipes, improve them, or add new ones. The more people contribute, the better it gets.

Why Should We Care?

- 📚 **Learning** – Read real-world code from experienced developers
- 💼 **Career** – Contributions are visible on your GitHub profile; employers notice
- 🤝 **Community** – Collaborate with developers worldwide from your laptop
- 💰 **Freedom** – Free tools, no expensive licenses
- 🛡️ **Quality** – Thousands of eyes catch bugs faster = more reliable & secure software

Open Source and the Internet

The internet **would not exist** without open source.

Layer	Open-Source Examples
Operating systems	Linux powers ~96% of top web servers
Web servers	Nginx, Apache
Languages	Python, JavaScript, PHP, Go, Rust
Databases	MySQL, PostgreSQL, MongoDB
Frameworks	React, Django, Flutter, Express
Infrastructure	Docker, Kubernetes, Terraform
Version control	Git itself (created by Linus Torvalds)

AI and Open Source

- 🧠 **AI frameworks are open source**: TensorFlow, PyTorch, Hugging Face
- 🤖 **Open models**: LLaMA, Mistral, Stable Diffusion – free to use and study
- ✨ **GitHub Copilot** was trained on open-source code – without it, AI coding assistants wouldn't exist
- 🔄 **AI helps open source back**: Copilot helps beginners contribute (exactly what we're doing today!)

> Open code trains better AI. Better AI helps more people write open code. They fuel each other.

How It All Connects to Cybersecurity

- 🛠️ **Security tools are open source**: Wireshark, Metasploit, Nmap, OWASP ZAP
- 🔍 **Transparency = trust**: Open code means no hidden backdoors
- ⚡ **Faster fixes**: Log4Shell (2021) was patched in days because Log4j was open source
- 🧠 **AI + security**: Threat detection, malware analysis – built on open-source AI frameworks
- 📱 **Today's app**: You're spreading cybersecurity awareness through a real, public app

> Open source, the internet, AI, and cybersecurity are all deeply connected. By contributing today, you're stepping into the ecosystem that powers modern technology.

Phase 0

Pre-Class Setup

⏱️ ~15 minutes

0.1 – Install Visual Studio Code

1. Go to **https://code.visualstudio.com/**
2. Click **Download for Windows**
3. Run the installer with default settings
4. ✅ Check **"Add to PATH"** on the Additional Tasks screen

> This lets you open VS Code from the terminal later

0.2 – Install Git for Windows

1. Go to **https://git-scm.com/download/win**
2. Download and run the installer
3. Accept all default settings
4. Installs `git` + **Git Bash** (a Linux-like terminal on Windows)

0.3 – Create a GitHub Account

1. Go to **https://github.com/join**
2. Sign up with a personal email
3. Verify your email
4. **Remember your username and password!**
5. Follow the college: **Kuwait-Technical-College** → click **Follow**

0.4 – Install GitHub Copilot Extension

1. Open VS Code
2. Press **Ctrl+Shift+X** → Extensions sidebar
3. Search for **"GitHub Copilot"**
4. Click **Install**
5. "GitHub Copilot Chat" installs automatically
6. **Sign in to GitHub** when prompted

> **Fallback**: Can't activate Copilot? Use Gemini free in a browser tab as an alternative.

✅ Setup Checklist

- ☐ VS Code installed and opens correctly
- ☐ Git installed (`git --version` shows a version number)
- ☐ GitHub account created and email verified
- ☐ GitHub Copilot extension installed and signed in

Phase 1

Orientation: VS Code and the Terminal

⏱️ ~20 minutes

1.1 – Tour of VS Code

Key areas to know:

Area	Shortcut	What It Does
File Explorer	`Ctrl+Shift+E`	Browse files and folders
Search	`Ctrl+Shift+F`	Search across all files
Source Control	`Ctrl+Shift+G`	Visual Git interface
Extensions	`Ctrl+Shift+X`	Install add-ons
Terminal	`Ctrl+` `	Built-in command line
Command Palette	`Ctrl+Shift+P`	Search any action

1.2 – What is a Terminal?

> Think of it as a **text-based version of File Explorer**. Instead of clicking to open folders, you **type commands**.

Open it with `Ctrl+` ` (backtick key, above Tab)

Developers use the terminal because it's **faster and more powerful** – many tools like Git are designed for it.

1.3 – Essential Terminal Commands

Command (Windows)	Command (macOS/Linux)	What It Does
`cd`	`pwd`	Shows current directory
`dir`	`ls`	Lists files and folders
`cd Desktop`	`cd Desktop`	Move into Desktop
`cd ..`	`cd ..`	Move up one level
`mkdir my-folder`	`mkdir my-folder`	Create a new folder
`cls`	`clear`	Clear the screen

> **Key concept**: You are always "inside" a folder – your **working directory**.

1.4 – Practice Exercise

```
cd Desktop
mkdir projects
cd projects
dir
```

You should now be inside an empty `projects` folder on your Desktop.

> 🤓 **Checkpoint**: You should see `Desktop\projects` as your current location.

Phase 2

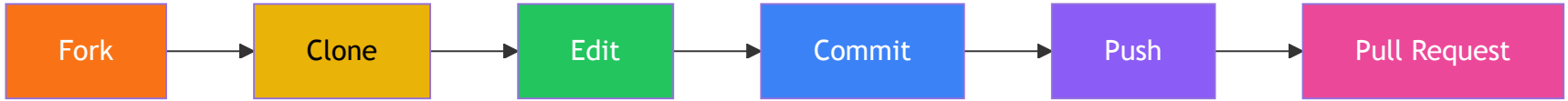
Git and GitHub: Fork and Clone

⏱️ ~25 minutes

2.1 – Key Concepts

Concept	Analogy
Git	"Track Changes" in Word, but for code
GitHub	Google Drive, but for code
Repository	A project folder tracked by Git
Fork	Your own copy of someone else's project
Clone	Downloading your fork to your computer
Commit	Saving a snapshot of your changes
Push	Uploading commits to GitHub

The Full Workflow



2.2 – Fork the Repository

1. Open **https://github.com/Kuwait-Technical-College/cybersecurity_zen_koans_app**
2. Click the **Fork** button (top-right)
3. Keep all defaults → Click **Create fork**

You now have your own copy at:

`https://github.com/YOUR-USERNAME/cybersecurity_zen_koans_app`

> "Nothing you do here affects the original until you submit a Pull Request."

2.3 – Clone to Your Computer

1. On your fork → Click green **Code** button
2. Ensure **HTTPS** is selected → Copy the URL
3. In VS Code terminal:

```
git clone https://github.com/YOUR-USERNAME/cybersecurity_zen_koans_app.git
cd cybersecurity_zen_koans_app
code .
```

> **Troubleshooting**: If `code .` doesn't work → **File → Open Folder**

2.4 – Configure Git Identity

One-time setup – Git needs to know who you are:

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
```

Use the **same email** you used for GitHub.

2.5 – Explore the Project

Navigate to: `src → assets → koans.json`

217 cybersecurity zen koans – short, thought-provoking sayings:

```
{
  "koanText": "The strongest firewall is useless against a user who clicks 'Allow'.",
  "technicalExplanation": "This koan highlights the critical importance of user behavior…",
  "uniqueCode": "7H93FT"
}
```

Field	Description
`koanText`	A short, wise cybersecurity saying
`technicalExplanation`	Detailed explanation of the concept
`uniqueCode`	Unique 6-character identifier

Phase 3

Using Copilot to Create a New Koan

⏱️ ~25 minutes

3.1 – Open Copilot Chat

1. Press **Ctrl+Shift+I** to open Copilot Chat
2. A chat panel opens – this is your AI assistant

> **What is GitHub Copilot?** A free AI assistant built into VS Code. Think of it as a knowledgeable coding partner you can chat with.

3.2 – Generate a Koan

Make sure `koans.json` is open, then ask Copilot:

> *"Generate a new cybersecurity zen koan in JSON format matching the structure in my open koans.json file. The koan should be about **phishing**. Include: koanText, technicalExplanation, and a unique 6-character alphanumeric uniqueCode."*

Pick your own topic!

Phishing	Ransomware	2FA	Social Engineering
Passwords	Wi-Fi Security	Updates	Data Privacy

3.3 – Review and Iterate

Ask yourself:

- Does the `koanText` sound **wise and thought-provoking**?
- Is the `technicalExplanation` **accurate**?
- Does it match the **style** of existing koans?

Refine with follow-up prompts:

- *"Make the koan shorter and more poetic"*
- *"Make the explanation more detailed"*
- *"Give me a different koan about the same topic"*

> 🧠 "AI is a tool, not a replacement for your brain. Always review what it generates."

3.4 – Add the Koan to koans.json

1. **Ctrl+End** to jump to the bottom
2. After the last `}`, before the `]`, add a comma
3. Paste your new koan

```
    "uniqueCode": "Q1PFVW"
  },
  {
    "koanText": "Your new koan text here",
    "technicalExplanation": "Your explanation here",
    "uniqueCode": "YOUR6C"
  }
]
```

⚠️ Watch for: missing comma, extra trailing comma, wrong field names. Red squiggly = fix it!

3.5 – Verify Your uniqueCode

Check that your code doesn't already exist:

Windows:

```
findstr "YOUR6C" src\assets\koans.json
```

macOS / Git Bash:

```
grep "YOUR6C" src/assets/koans.json
```

- **No output** → ✅ Code is unique
- **A line appears** → ❌ Generate a different code

Bonus – count total koans:

```
grep -c "uniqueCode" src/assets/koans.json
# Should print 218 after adding yours
```

Phase 4

Commit, Push, and Pull Request

⏱️ ~25 minutes

4.1 – Check What Changed

```
git status
```

```
modified:   src/assets/koans.json
```

See the exact changes:

```
git diff src/assets/koans.json
```

- Lines in **green** (`+`) = added
- Lines in **red** (`-`) = removed

> 💡 **Visual alternative**: `Ctrl+Shift+G` → Source Control → click the file for a side-by-side diff

4.2 – Stage Your Changes

```
git add src/assets/koans.json
```

> **What is staging?** Like putting items in a box before shipping. You choose exactly which changes go into this commit.

4.3 – Commit Your Changes

```
git commit -m "Add new cybersecurity koan about phishing"
```

> **What is a commit?** A save point in a video game. It records the exact state of your files. The `-m` flag adds a description message.

4.4 – Push to GitHub

```
git push origin main
```

- `origin` = your GitHub fork (where you cloned from)
- `main` = the branch name

> **Authentication**: Windows may open a browser to sign in. If you get errors, you may need a Personal Access Token.

4.5 – Verify on GitHub

1. Open your fork: `github.com/YOUR-USERNAME/cybersecurity_zen_koans_app`
2. See your commit at the top
3. Click `src/assets/koans.json` → scroll to bottom → see your koan

4.6 – Create a Pull Request

1. On your fork, see: *"This branch is 1 commit ahead"*
2. Click **Contribute** → **Open pull request**
3. **Title**: `Add new koan: [brief description]`
4. **Description**: 1–2 sentences about your koan topic
5. Click **Create pull request**

🎉 Congratulations!

✅ Used a professional code editor (**VS Code**)

✅ Navigated the **terminal** like a developer

✅ Used **AI** (GitHub Copilot) to create content

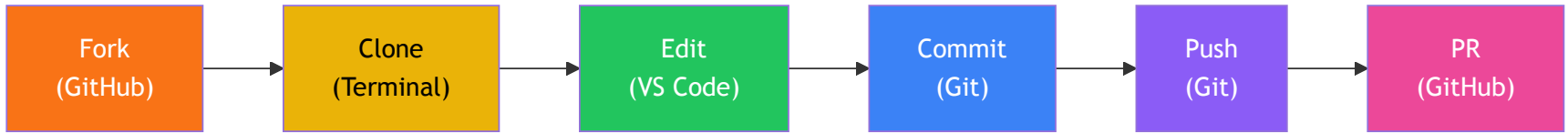
✅ Used **Git** to track and submit changes

✅ Contributed to a **real open-source project**

Your Pull Request is now live on the project's PR page! 🚀

> Once merged, a **new version of the app** ships to the **App Store** & **Play Store** – with your koan in it! 📱

The Workflow You Learned



Keep Learning

Resource	Link
VS Code Docs	code.visualstudio.com/docs
Git Basics (free book)	git-scm.com/book
GitHub Skills	skills.github.com
GitHub Copilot Docs	docs.github.com/copilot

Bonus Challenges

- Add **2 or more koans** in a single PR
- **Review** another student's Pull Request
- **Explore** the repo – how does `koans.json` get loaded?

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Thank You

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Happy contributing! 🎉